

Twish Sanghvi

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SUMMARY

Research Assistant experienced in semiconductor fabrication and device simulation. Skilled in SILVACO TCAD, photolithography, and cleanroom processes, currently fabricating AlN FinFETs. Strong in device physics and process optimization. Seeking a Summer 2026 Process Development Engineering Internship in MEMS and advanced semiconductor manufacturing.

EDUCATION

B.S.E., Electrical Engineering

Arizona State University, Tempe, AZ

May 2027

3.96 GPA

TECHNICAL SKILLS

Design and Simulation Tools: SILVACO TCAD, Process Optimization, Semiconductor Device Physics

Fabrication and Lab Techniques: Photolithography (spin-coating, mask alignment, UV exposure, development), Critical Dimension Control, Overlay Metrology, Etching (wet & dry), Thin Film Deposition

Programming: C++, Java, MATLAB

WORK EXPERIENCE

Research Assistant – Semiconductor Simulation and Fabrication

February 2025 – present

Arizona State University

Researching ultra-wide bandgap AlN FinFETs to advance high-power, high-frequency device performance and enhance reliability in extreme operating conditions.

- Executed **30+ complete lithography runs** with **sub-micron alignment ($\pm 0.2 \mu\text{m}$)** using Heidelberg MLA150 and OAI 808 Aligner, enabling precise AlN FinFET patterning.
- Implemented **inline overlay and CD metrology** across **10+ wafers**, reducing rework cycles by **25 %** and improving overall process repeatability.
- Collaborated with process engineers to lower **defect density by 15 %**, directly improving wafer yield and uniformity.
- Performed **thin-film characterization** (SEM, ellipsometry) to correlate surface roughness and refractive index with deposition conditions, improving film uniformity by **10 %**.
- Applied **MEMS-grade fabrication techniques**—controlled etch rates $< 50 \text{ nm/min}$ and film deposition $< 100 \text{ nm}$ precision—to produce **robust, high-aspect-ratio FinFET prototypes**.

Tutor – Academic Support Network

January 2025 – present

Arizona State University

- Tutored undergraduate students in **circuits, digital design, electromagnetics, and signals & systems**, reinforcing key concepts through examples and problem-solving strategies.
- Supported 200+ students per semester, leading to measurable improvements in course performance and confidence.

PROJECTS

$\beta\text{-Ga}_2\text{O}_3$ Schottky Diode Modelling

Spring 2025 – Summer 2025

Arizona State University

- Designed **vertical $\beta\text{-Ga}_2\text{O}_3$ diodes** with doping profiles from 1×10^{16} to $1 \times 10^{18} \text{ cm}^{-3}$, achieving **$> 20 \%$ improvement** in breakdown voltage.
- Conducted **50 + SILVACO TCAD simulations** to study electric-field distribution and I–V behavior with $< 5 \%$ deviation from theoretical values.
- Optimized **mask geometry and doping gradients**, enhancing **device manufacturability and simulation throughput by 30 %**.

16-Bit Microprocessor Design in Verilog

Spring 2024

Arizona State University

- Designed and implemented a 16-bit microprocessor** from scratch in **Verilog**, integrating ALUs, registers, control units, data buses, and memory interfaces.
- Utilized simulation tools to verify pipeline performance, debug control signals, and validate instruction execution, demonstrating strong **digital design and hardware description language (HDL)** skills.
- Built and tested modular components (arithmetic logic units, register files, and instruction decoding) to ensure functional correctness and timing reliability.